Chapter 2
Bits, Data Types, and Operations

陈俊仕
cjuns@ustc.edu.cn
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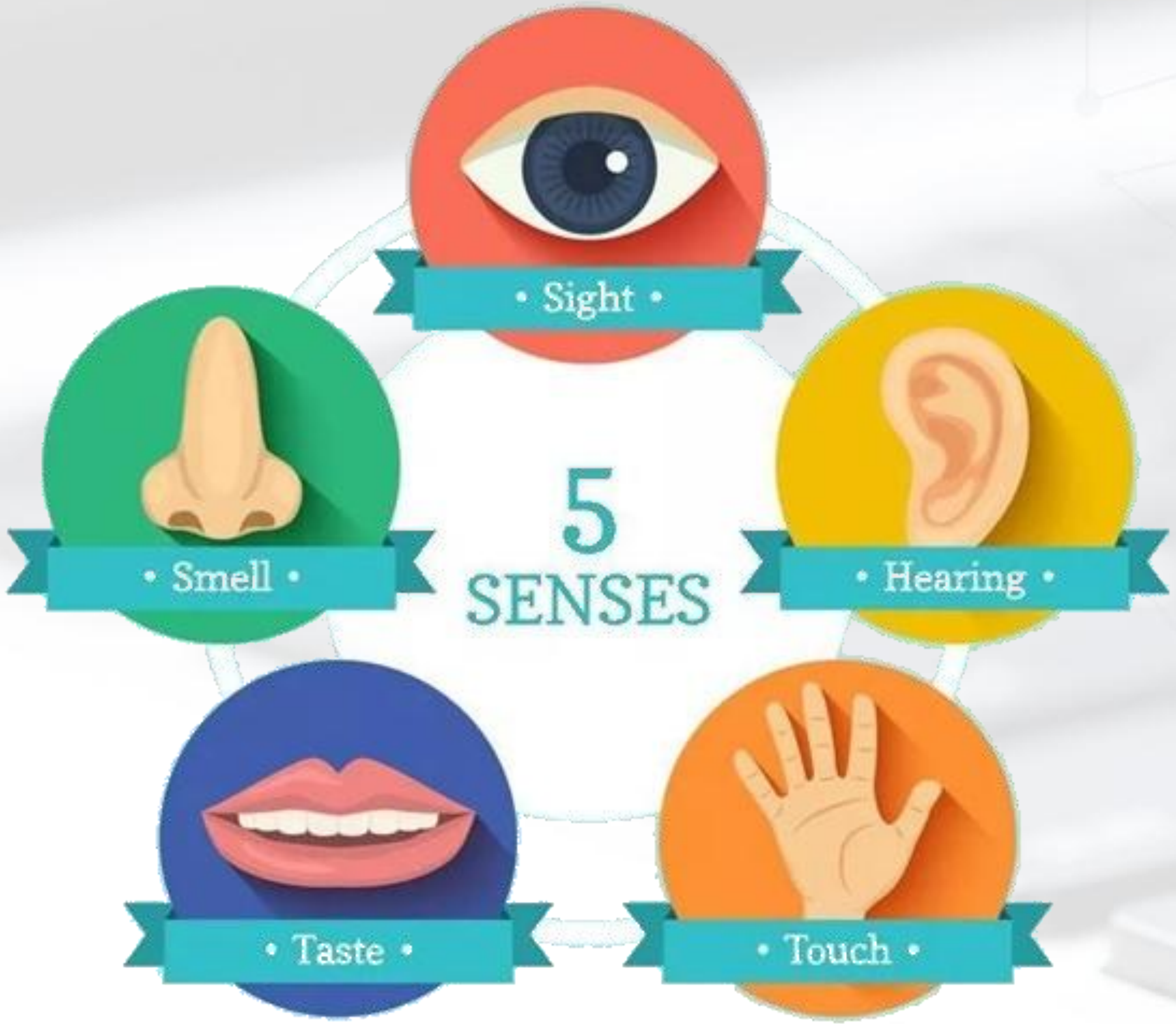
计算机科学与技术学院
School of Computer Science and Technology

Outline

Outline

5 Senses of Human

- **Sight**
  - Image, picture, photo, video, …
- **Hearing**
  - Sound, voice, speech, music, …
- **Touch**
  - Shape, soft, hard, hurt, numb, …
- **Taste**
  - Sour, sweet, bitter, spicy, salty, …
- **Smell**
  - Sweet, smelly, …



to record by number, data, words, symbols, text, language, ……

What kinds of information do we need to represent?

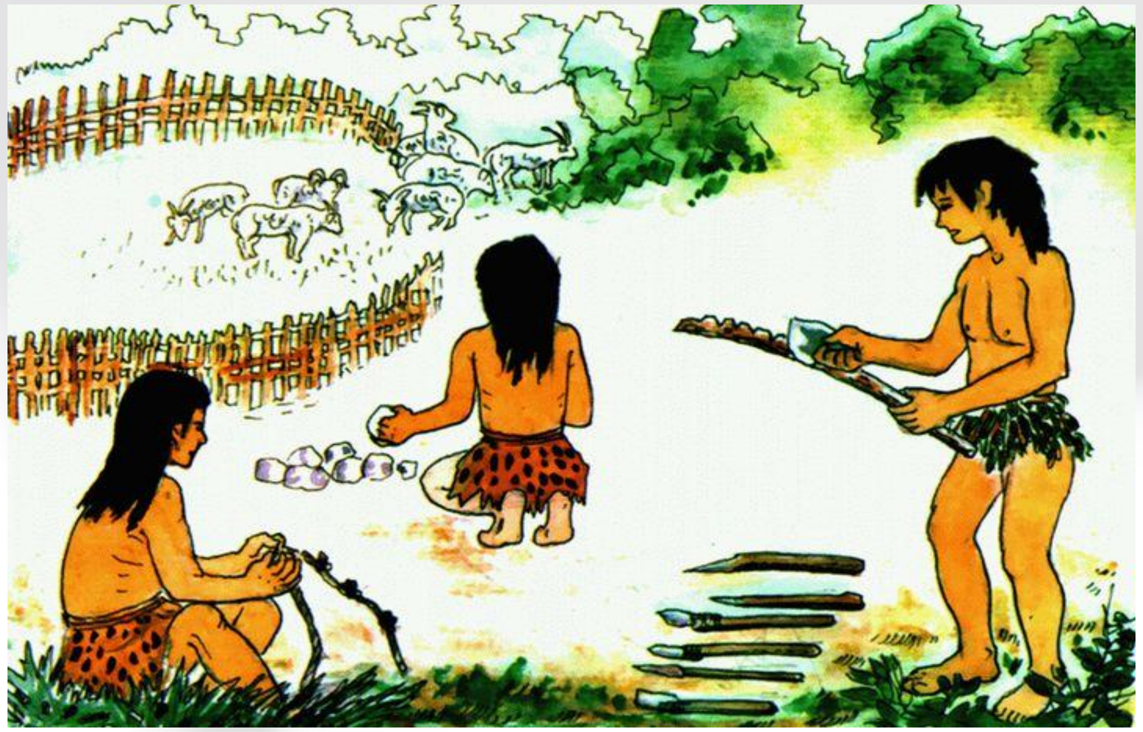
Kinds of Information

- **Numbers** – natural number, integers, positive/negative integers, integers/decimals, real, complex, rational, irrational, signed, unsigned, floating point, …
- **Text** – characters, strings, …
- **Logical** – true, false
- **Images** – pixels, colors, shapes, …
- **Sound** – sound of talk, sound of sing, …
- **Video** – a series of images
- **Instructions** – plus(+), minus(-), times (*) ,divided by(/) , …
- …

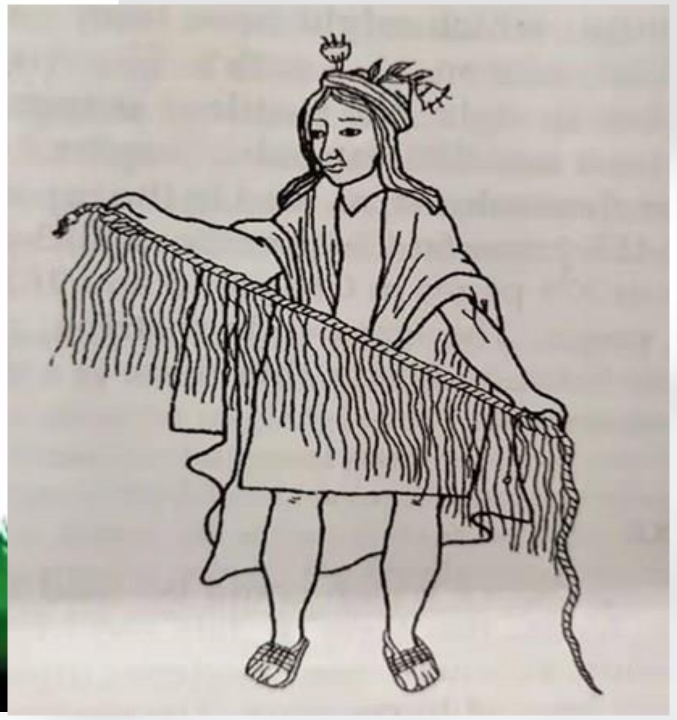
Data type: *representation* and *operations* within the computer

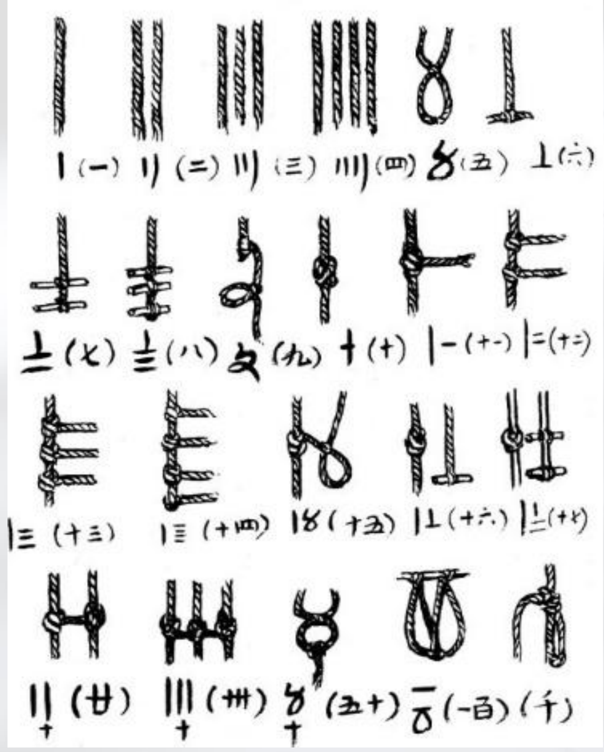
We' ll start with **numbers**…

Number Notation



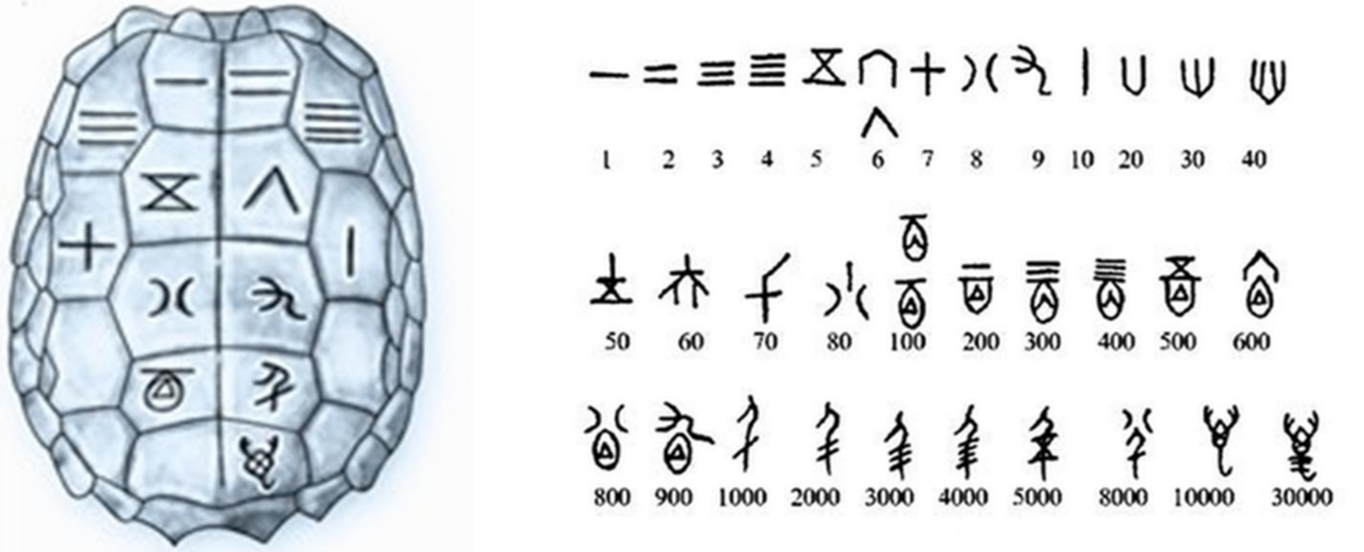
Counting stone(石头)





Counting rod(算筹)

Knotting(结绳)



Inscriptions on oracle bones
(甲骨文上刻字)

Number Notation

- Non-positional notation(like to counting rod)
  - Could represent a number ("5") with a string of ones ("11111") problems?

5

V

11111

Number Notation

Weighted positional notation

- decimal numbers(denary numbers): "329"
- "3" is worth 300, because of its position(with place value 100),
- while "9" is only worth 9, because of its position(with place value 1)

329

10^2 10^1 10^0

$\boxed{3x100 + 2x10 + 9x1 = 329}$

Denary numbers
base is 10,
place value according its position

Denary numbers - base ten

- $(5346)_{10}$

5346

Available digit	0, 1, 2, 3, 4, 5, 6, 7, 8, 9			
Place value	$10^3=1000$	$10^2=100$	$10^1=10$	$10^0=1$
Digit	5	3	4	6
Product of digit and place value	5x1000=5000	3x100=300	4x10=40	6x1=6

$(5346)_{10}$ = 5x1000+3x100+4x10+6x1

How do we represent data in a computer?

Great Idea from Ancient Chinese Philosophy

All things come into being, all things come into nothing

天下万物生于有，有生于无　——《老子·四十章》

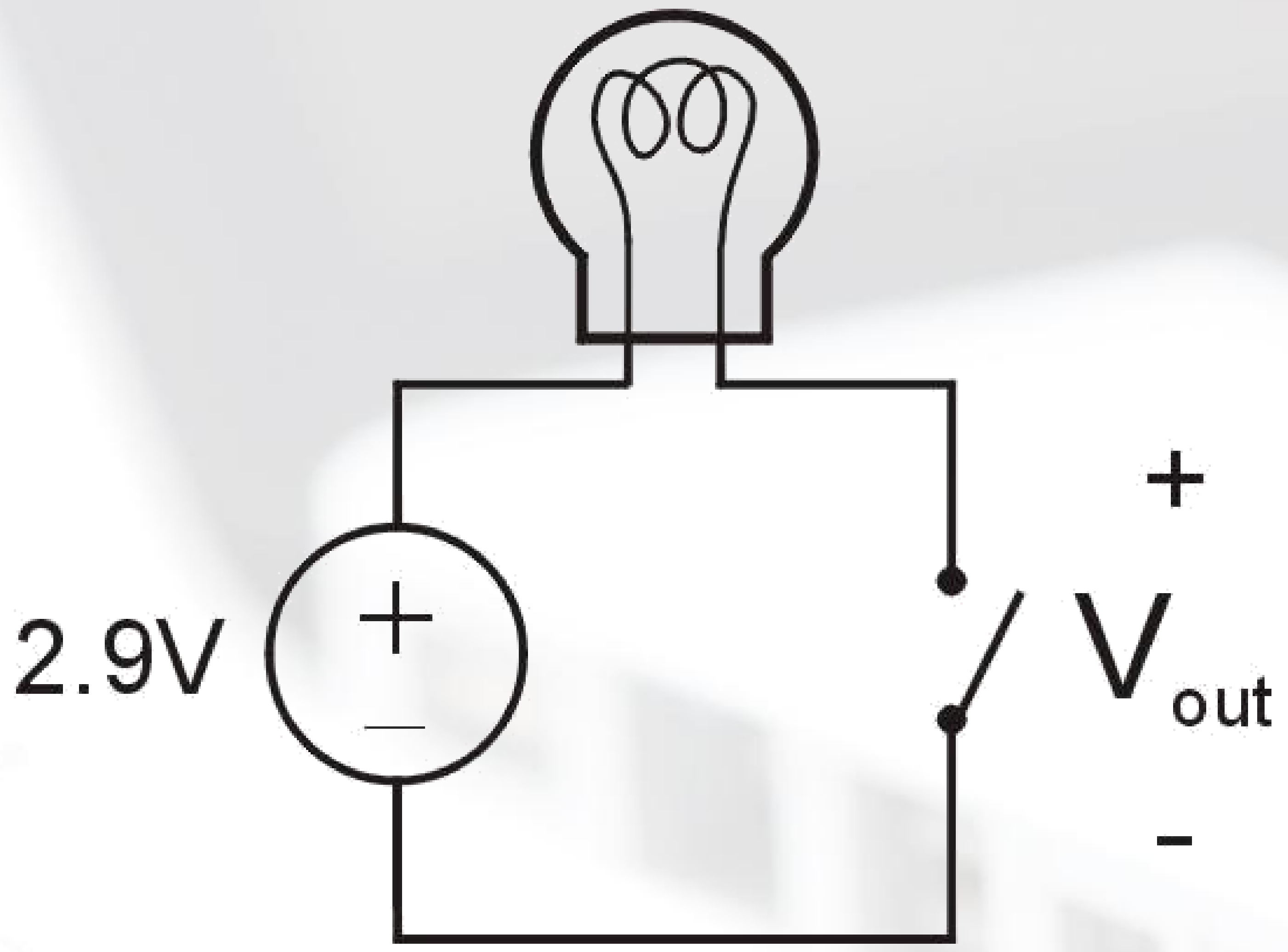
《易经》

太极生两仪，
两仪生四象，
四象生八卦，
八卦演万物。

How do we represent data in a computer?

- At the lowest level, a computer is an electronic machine.
  - works by controlling *the flow of electrons*
- Easy to recognize two conditions:
  - presence of a voltage – we'll call this state "1"
  - absence of a voltage – we'll call this state "0"
- Could base state on *value* of voltage, but control and detection circuits more complex.
  - compare turning on a light switch to measuring or regulating voltage
- We' ll see examples of these circuits in the next chapter.

Simple Switch Circuit



Switch open:

- No current through circuit
- Light is off
- V_{out} is +2.9V

Switch closed:

- Short circuit across switch
- Current flows
- Light is on
- V_{out} is 0V

Switch-based circuits can easily represent two states:
on/off, open/closed, voltage/no voltage.

Computer is a binary digital system

Digital system:
- finite number of symbols

Binary (base two) system:
- has two states: 0 and 1

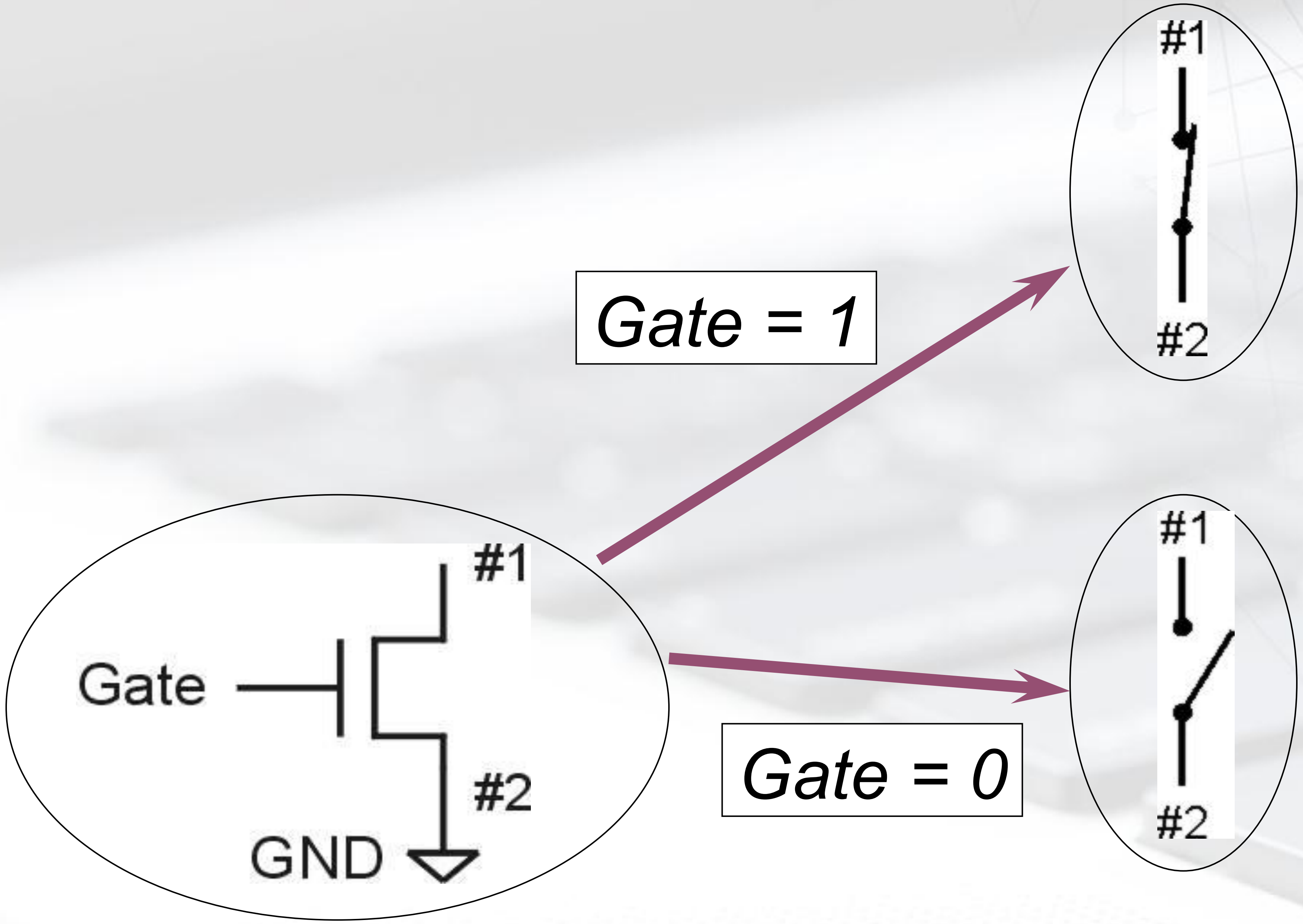
Basic unit of information is the *binary digit*, or *bit*.

Values with more than two states require multiple bits.

- A collection of two bits has four possible states: 00, 01, 10, 11
- A collection of three bits has eight possible states: 000, 001, 010, 011, 100, 101, 110, 111
- ***A collection of n bits has 2^n possible states.***

N-type MOS Transistor

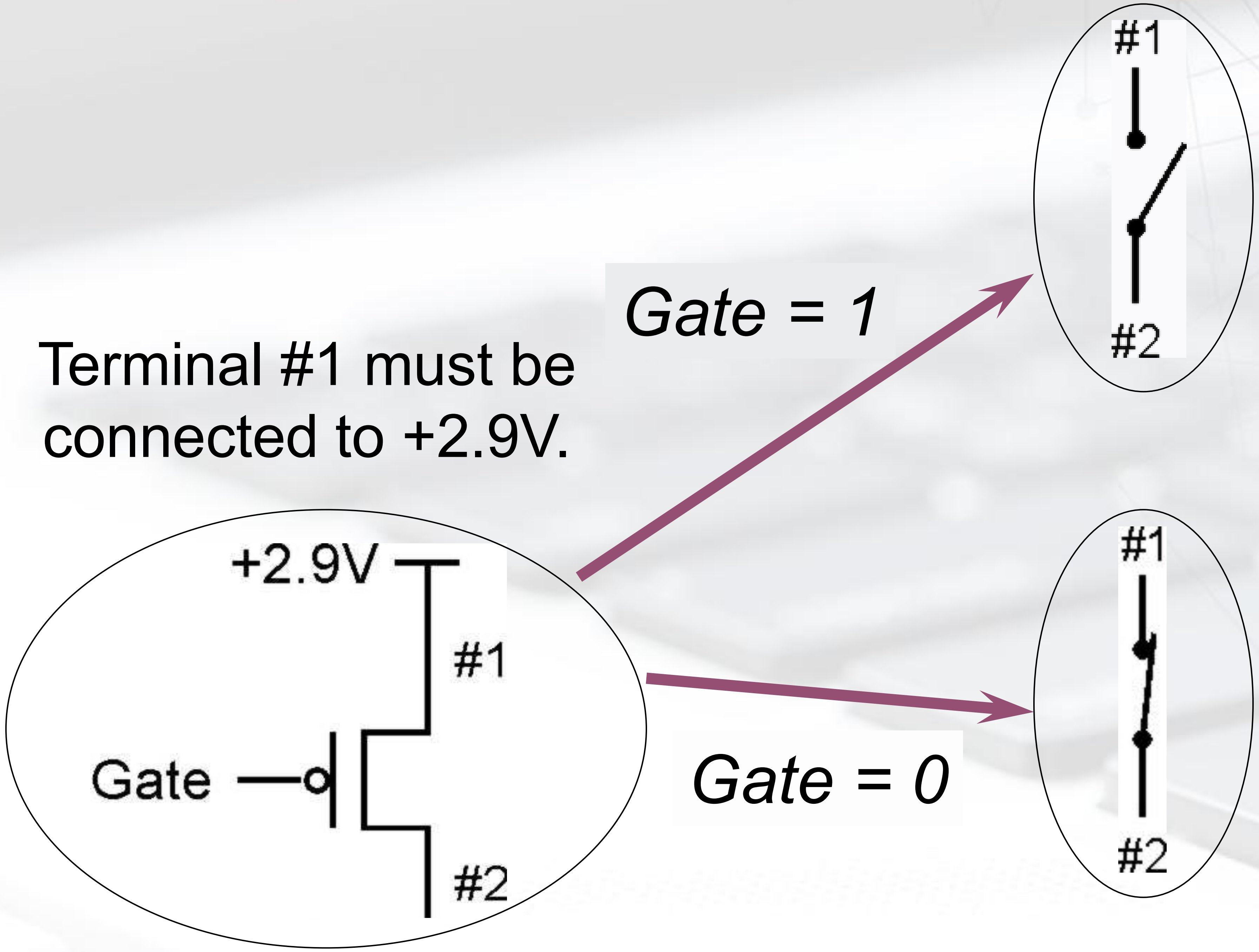
- **MOS = Metal Oxide Semiconductor**
  - two types: N-type and P-type
- **N-type**
  - when Gate has positive voltage, short circuit between #1 and #2 (switch closed)
  - when Gate has zero voltage, open circuit between #1 and #2 (switch open)



Terminal #2 must be connected to GND (0V).

P-type MOS Transistor

- **P-type is *complementary* to N-type**
  - when Gate has positive voltage, open circuit between #1 and #2 (switch open)
  - when Gate has zero voltage, short circuit between #1 and #2 (switch closed)



Logic Gates

- Use switch behavior of MOS transistors to implement logical functions: **AND, OR, NOT.**
- Digital symbols:
  - recall that we assign a range of analog voltages to each digital (logic) symbol

  - assignment of voltage ranges depends on electrical properties of transistors being used
  - typical values for "1": +5V, +3.3V, +2.9V, +1.1V for purposes of illustration, we'll use +2.9V

Binary numbers - base two

- $(101110)_2$

10 1110

Available digit	0, 1					
Place value	$2^5=32$	$2^4=16$	$2^3=8$	$2^2=4$	$2^1=2$	$2^0=1$
Digit	1	0	1	1	1	0
Product of digit and place value	32	0	8	4	2	0

$(101110)_2 = 1\text{x}32+0\text{x}16+1\text{x}8+1\text{x}4+1\text{x}2+0\text{x}1 = (46)_{10}$

$(11110100)_2 = 1\text{x}128+1\text{x}64+1\text{x}32+1\text{x}16+0\text{x}8+1\text{x}4+0\text{x}2+0\text{x}1=(244)_{10}$

$(2790)_{10} = (\quad ? \quad)_2$

$(5346)_{10} = (\quad ? \quad)_2$

Within the Computer: Everything is a Number.

■ Numbers within the Computer

- Base 10 #s: Dec(imal)
  - Digits: 0,1,2,3,4,5,6,7,8,9
- Base 2 #s: Bin(ary)
  - Digits: 0,1
- Base 8 #s: Oct(al)
  - Digits: 0,1,2,3,4,5,6,7
- Base 16 #s: Hex(adecimal)
  - Digits: 0,1,2,3,4,5,6,7,8,9,A,B,C,D,E,F

Dec(imal)	Hex(adecimal)	Oct(al)	Bin(ary)
00	0	00	0000
01	1	01	0001
02	2	02	0010
03	3	03	0011
04	4	04	0100
05	5	05	0101
06	6	06	0110
07	7	07	0111
08	8	10	1000
09	9	11	1001
10	A	12	1010
11	B	13	1011
12	C	14	1100
13	D	15	1101
14	E	16	1110
15	F	17	1111

Hexadecimal Notation

- **It is often convenient to write binary (base-2) numbers as hexadecimal (base-16) numbers instead.**
  - `fewer digits -- four bits per hex digit`
  - `less error prone -- easy  to corrupt long string of 1's and 0's`

Binary	Hex	Decimal	Binary	Hex	Decimal
0000	0	0	1000	8	8
0001	1	1	1001	9	9
0010	2	2	1010	A	10
0011	3	3	1011	B	11
0100	4	4	1100	C	12
0101	5	5	1101	D	13
0110	6	6	1110	E	14
0111	7	7	1111	F	15

011101010001111010011010111

Converting from Binary to Hexadecimal

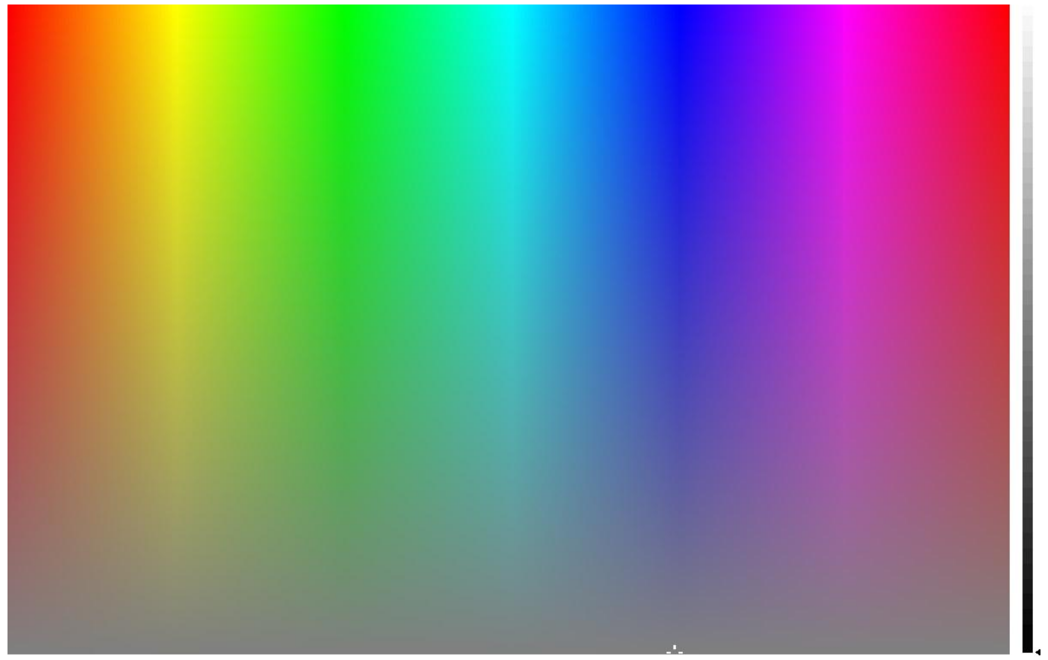
- **Every four bits is a hex digit.**
  - `start grouping from right-hand side`

```
 011101010001111010011010111
   3   A   8   F   4   D   7
```

This is not a new machine representation, just a convenient way to write the number.

BIG IDEA：Bits can represent anything!!!

- **Characters?**
  - 26 letters ⇒ 5 bits ($2^5 = 32$)
  - upper/lower case + punctuation（符号） ⇒ 7 bits (in 8)(“ASCII”)
  - standard code to cover all the world’s languages ⇒ 8,16,32 bits (“Unicode”) www.unicode.com
- **Logical values?**
  - 0 → False, 1 → True
- **colors ?**
  - Ex: Red(00),Green(01),Blue(11)
- **locations / addresses?**
- **commands?**



MEMORIZE: N bits ⇔ at most 2^N things

Within the Computer: Everything is a Number.

- **Bit(BInary digiT)**
  - `1Bits=2things;`
  - `2Bits=4things;`
  - `4Bits=16things;`
  - `8Bits=256things`
  - ……
- **Byte**
  - `1Byte=8Bits`
  - `A byte is 8 bits`
- **But numbers usually stored with a fixed size**
  - `8-bit bytes;`
  - `16-bit half words;`
  - `32-bit words;`
  - `64-bit double words, ...`
  - `And there are really only two primitive "numbers": 0 and 1 is a "bit"`

Outline

Unsigned Integers

Weighted positional notation

- like decimal numbers: “329”
- “3” is worth 300, because of its position, while “9” is only worth 9

329

10^2 10^1 10^0

$\boxed{3x100 + 2x10 + 9x1 = 329}$

MSB *most significant*

101

LSB *least significant*

2^2 2^1 2^0

$\boxed{1x4 + 0x2 + 1x1 = 5}$

Unsigned Integers

- An *n*-bit unsigned integer represents 2^n values: from 0 to 2^n-1.

2^2	2^1	2^0	
0	0	0	0
0	0	1	1
0	1	0	2
0	1	1	3
1	0	0	4
1	0	1	5
1	1	0	6
1	1	1	7

Unsigned Binary Arithmetic

■ Base-2 addition – just like base-10!

- add from right to left, propagating carry

carry

```
  10010       10010       1111
+  1001     +  1011     +    1
  11011       11101      10000

              10111
            +   111
```

- Subtraction, multiplication, division,…

Signed Integers

- With n bits, we have 2^n distinct values.
  - assign about half to positive integers (1 through 2^{n-1}) and about half to negative (-2^{n-1} through -1)
  - that leaves two values: one for 0, and one extra
- Positive integers
  - just like unsigned – zero in Most Significant (MS) bit
    00101 = 5
- Negative integers
  - sign-magnitude (原码) – set top bit to show negative, other bits are the same as unsigned
    10101 = -5
  - one's complement (反码) – flip every bit to represent negative
    11010 = -5
  - in either case, MS bit indicates sign: 0=positive, 1=negative

Three representations of signed integers

Representation					Value Represented		
					Signed Magnitude	1's Complement	
0	0	0	0	0	0	0	
0	0	0	0	1	1	1	
0	0	0	1	0	2	2	
0	0	0	1	1	3	3	
0	0	1	0	0	4	4	
0	0	1	0	1	5	5	
0	0	1	1	0	6	6	
0	0	1	1	1	7	7	
0	1	0	0	0	8	8	
0	1	0	0	1	9	9	
0	1	0	1	0	10	10	
0	1	0	1	1	11	11	
0	1	1	0	0	12	12	
0	1	1	0	1	13	13	
0	1	1	1	0	14	14	
0	1	1	1	1	15	15	

Representation					Value Represented		
					Signed Magnitude	1's Complement	
1	0	0	0	0	—0	—15	
1	0	0	0	1	—1	—14	
1	0	0	1	0	—2	—13	
1	0	0	1	1	—3	—12	
1	0	1	0	0	—4	—11	
1	0	1	0	1	—5	—10	
1	0	1	1	0	—6	—9	
1	0	1	1	1	—7	—8	
1	1	0	0	0	—8	—7	
1	1	0	0	1	—9	—6	
1	1	0	1	0	—10	—5	
1	1	0	1	1	—11	—4	
1	1	1	0	0	—12	—3	
1	1	1	0	1	—13	—2	
1	1	1	1	0	—14	—1	
1	1	1	1	1	—15	—0	

Signed Magnitude：

5 - 5 = 5 + (-5) =-10

```
    00101     (5)
+   10101      (-5)
    11010   (-10)
```

1's Complement：

5 - 5 = 5 + (-5) =- 0

```
    00101     (5)
+   11010    (-5)
    11111    (-0)
```

Outline

1. How do we represent information in a computer?
2. Integer Data Types
3. 2’ Complement Integers
4. Binary-Decimal Conversion
5. OComplementperations on Bits: Arithmetic and Logical
6. Other Representation

Modular Arithmetic and Signed Integer Representation

- *a* mod *m* –the residue of *a* divided by *m*

  *a=q*m+ r ;   0<=r<m ;   r=a mod m*

  - Example: 14 mod 4=2 due to 14=3*4+2; 2 is the residue of 14 mod 4
  - -3 mod 8=5 due to -3=(-1)*8+5;
- Registers in a computer have *finite length*
  - e.g. length of n bits, or 2^n distinct numbers can be represented
  - Addition, subtraction in a computer is actually modular arithmetic.
- For unsigned numbers
  - Example: length of 3 bits-000, 001, 010, 011, 100, 101, 110, 111
  - In decimal (increase by 1, finite representations)

0 mod 8=0;	1 mod 8=1;	2 mod 8=2;	3 mod 8=3;	4 mod 8=4;	5 mod 8=5;
000	001	010	011	100	101

6 mod 8 =6;	7 mod 8=7;	8 mod 8=0;	9 mod 8=1;	10 mod 8=2;	…
110	111	000	001	010	

- **How to representing -4,-3,-2,-1,0,1,2,3 in 3 bits?**

  keep 0,1,2,3 unchanged, how to represent -4,-3,-2,-1 and get arithmetic works?

Modular Arithmetic and Signed Integer Representation

- **The property of Modular Arithmetic: If 0<=a,b<m, then**
  - (a+b) mod m=(a mod m+ b mod m) mod m;
    (5+7)mod 8= 12 mod 8=4;
  - (a-b)mod m=(a+m-b)mod m=(a + (m-b)) mod m
    (7-5) mod 8=(7+(8-5))mod 8=(7+3)mod 8=2

- **(m-b)mod m=(-b) mod m; (m-b) is the complement of (-b) when reduced modulo m.**
  -3 mod 8 = (8-3) mod 8 = 5 mod 8 → -3 = 5 mod 8: *to support modular arithmetic*, signed -3 can be represented as unsigned 5. i.e., -3: 101
  Similarly, -1=7 mod 8→-1:111; -2=6 mod 8→-2:110; -4=4 mod 8→ -4:100

- **Conclusion: a (signed) negative number (e.g., –b) can be represented in binary by its (unsigned) positive complement mod m (e.g., m-b).**

- **Since (m-b) +b =m mod m=0, to represent –b, -b=m-b=((m-1)-b)+1, 1-0=1, 1-1=0**
  - 1) represent b in binary,
  - 2) flip each bit of b and add 1 to obtain the representation of (m-b), such that the sum of both representations is 0.

Two’s Complement Representation

- **If number is positive or zero,**
  - normal binary representation, zeroes in upper bit(s)
- **If number is negative,**
  - start with positive number
  - flip every bit (i.e., take the one’s complement)
  - then add one
- **This representation makes the hardware simple!**

	00101	(5)
	11010	(1's comp)
+	1	
	11011	(-5)

	01001	(9)
		(1's comp)
+	1	
		(-9)

Two’ s Complement

- **Problems with sign-magnitude and 1’ s complement**
  - `two representations of zero (+0 and -0)`
  - `arithmetic circuits are complex`
    - **How to add two sign-magnitude numbers?**
      - e.g., try 2 + (-3)
    - **How to add two one’s complement numbers?**
      - e.g., try 4 + (-3)
- ***Two’ s complement*** **representation developed to make circuits easy for arithmetic.**
  - `for each positive number (X), assign value to its negative (-X), such that X + (-X) = 0 with "normal" addition, ignoring carry out`

00101	(5)	01001	(9)
+ 11011	(-5)	+ ______	(-9)
00000	(0)	00000	(0)

Two’ s Complement Shortcut

- To take the two’ s complement of a number:
  - copy bits from right to left until (and including) the first “1”
  - flip remaining bits to the left

```
  011010000                       0110 10000
  100101111   (1’s comp)   (flip)  |    | (copy)
+         1
  100110000                       1001 10000
```

Two’ s Complement Signed Integers

- MS bit is sign bit – it has weight -2^{n-1}.
- Range of an n-bit number: -2^{n-1} through $2^{n-1} - 1$.
  - The most negative number (-2^{n-1}) has no positive counterpart.

-2^3	2^2	2^1	2^0	
0	0	0	0	0
0	0	0	1	1
0	0	1	0	2
0	0	1	1	3
0	1	0	0	4
0	1	0	1	5
0	1	1	0	6
0	1	1	1	7

-2^3	2^2	2^1	2^0	
1	0	0	0	-8
1	0	0	1	-7
1	0	1	0	-6
1	0	1	1	-5
1	1	0	0	-4
1	1	0	1	-3
1	1	1	0	-2
1	1	1	1	-1

Three representations of signed integers

Representation	Value Represented: Signed Magnitude	Value Represented: 1's Complement	Value Represented: 2's Complement
0 0 0 0 0	0	0	0
0 0 0 0 1	1	1	1
0 0 0 1 0	2	2	2
0 0 0 1 1	3	3	3
0 0 1 0 0	4	4	4
0 0 1 0 1	5	5	5
0 0 1 1 0	6	6	6
0 0 1 1 1	7	7	7
0 1 0 0 0	8	8	8
0 1 0 0 1	9	9	9
0 1 0 1 0	10	10	10
0 1 0 1 1	11	11	11
0 1 1 0 0	12	12	12
0 1 1 0 1	13	13	13
0 1 1 1 0	14	14	14
0 1 1 1 1	15	15	15

Representation	Value Represented: Signed Magnitude	Value Represented: 1's Complement	Value Represented: 2's Complement
1 0 0 0 0	—0	—15	—16
1 0 0 0 1	—1	—14	—15
1 0 0 1 0	—2	—13	—14
1 0 0 1 1	—3	—12	—13
1 0 1 0 0	—4	—11	—12
1 0 1 0 1	—5	—10	—11
1 0 1 1 0	—6	—9	—10
1 0 1 1 1	—7	—8	—9
1 1 0 0 0	—8	—7	—8
1 1 0 0 1	—9	—6	—7
1 1 0 1 0	—10	—5	—6
1 1 0 1 1	—11	—4	—5
1 1 1 0 0	—12	—3	—4
1 1 1 0 1	—13	—2	—3
1 1 1 1 0	—14	—1	—2
1 1 1 1 1	—15	—0	—1

Signed Magnitude:

5 - 5 = 5 + (-5) =-10

```
    00101     (5)
+   10101      (-5)
    11010   (-10)
```

1's Complement:

5 - 5 = 5 + (-5) =- 0

```
    00101      (5)
+   11010     (-5)
    11111     (-0)
```

2's Complement:

5 - 5 = 5 + (-5) = 0

```
    00101      (5)
+   11011      (-5)
    00000      (0)
```

Question

- **Why is 1's complement called 1's complement?**

  1-0=1; 1-1=0

- **Why is 2's complement called 2's complement?**

  $$-b = 2^n - b$$

Q&A

- Suppose we had a 5-bit word. What integers can be represented in two’ s complement?
  A. -32~+31
  B. 0~+31
  C. -16~+15
  D. -15~+16
- Suppose we had an 8-bit word. What integers can be represented in two’ s complement?
- Suppose we had a 16-bit word. What integers can be represented in two’ s complement?
- Suppose we had a 32-bit word. What integers can be represented in two’ s complement?

Outline

Converting Binary (2’s C) to Decimal

1. If leading bit is one, take two’s complement to get a positive number.
2. Add powers of 2 that have “1” in the corresponding bit positions.
3. If original number was negative, add a minus sign.

$$X = 01101000_{two}$$
$$= 2^6+2^5+2^3 = 64+32+8$$
$$= 104_{ten}$$

Assuming 8-bit 2’s complement numbers.

n	2^n
0	1
1	2
2	4
3	8
4	16
5	32
6	64
7	128
8	256
9	512
10	1024

More Examples

$$
\begin{aligned}
X &= 00100111_{two} \\
&= 2^5+2^2+2^1+2^0 = 32+4+2+1 \\
&= 39_{ten}
\end{aligned}
$$

$$
\begin{aligned}
X &= 11100110_{two} \\
-X &= 00011010 \\
&= 2^4+2^3+2^1 = 16+8+2 \\
&= 26_{ten} \\
X &= -26_{ten}
\end{aligned}
$$

Assuming 8-bit 2's complement numbers.

n	2^n
0	1
1	2
2	4
3	8
4	16
5	32
6	64
7	128
8	256
9	512
10	1024

Converting Decimal to Binary (2' s C)

First Method: *Division*

1. Divide by two – remainder is least significant bit.
2. Keep dividing by two until answer is zero, writing remainders from right to left.
3. Append a zero as the MS bit; if original number negative, take two's complement.

$X = 104_{ten}$

104/2	=	52 r0	*bit 0*
52/2	=	26 r0	*bit 1*
26/2	=	13 r0	*bit 2*
13/2	=	6 r1	*bit 3*
6/2	=	3 r0	*bit 4*
3/2	=	1 r1	*bit 5*
1/2	=	0 r1	*bit 6*

$X = 01101000_{two}$

n	2^n
0	1
1	2
2	4
3	8
4	16
5	32
6	64
7	128
8	256
9	512
10	1024

Converting Decimal to Binary (2’ s C)

Second Method: *Subtract Powers of Two*

1. Change to positive decimal number.
2. Subtract largest power of two less than or equal to number.
3. Put a one in the corresponding bit position.
4. Keep subtracting until result is zero.
5. Append a zero as MS bit; if original was negative, take two’s complement.

n	2^n
0	1
1	2
2	4
3	8
4	16
5	32
6	64
7	128
8	256
9	512
10	1024

$X = 104_{ten}$

104 - 64 =	40	*bit 6*
40 - 32 =	8	*bit 5*
8 - 8 =	0	*bit 3*

$X = 01101000_{two}$

Outline

Operations: Arithmetic and Logical

- Recall: a data type includes *representation* and *operations*.
- We now have a good representation for signed integers, so let’ s look at some arithmetic operations:
  - Addition
  - Subtraction
  - Sign Extension ！！！
- We’ ll also look at **overflow** conditions for addition.
- Multiplication, division, etc., can be built from these basic operations.
- Logical operations are also useful:
  - AND, OR, NOT

Addition

- **As we’ ve discussed, 2’ s comp. addition is just binary addition.**
  - assume all integers have the same number of bits
  - ignore carry out
  - for now, assume that sum fits in n-bit 2’s comp. representation

01101000 (104)	11110110 (-10)
+ 11110000 (-16)	+ (-9)
01011000 (88)	(-19)

Assuming 8-bit 2’s complement numbers.

Subtraction

Negate subtrahend (2nd no.) and add.

- assume all integers have the same number of bits
- ignore carry out
- for now, assume that difference fits in n-bit 2's comp. representation

```
   01101000  (104)          11110110 (-10)
-  00010000   (16)       -              (-9)
___________________      _________________

   01101000  (104)          11110110 (-10)
+  11110000  (-16)       +               (9)
___________________      _________________
   01011000   (88)                   (-1)
```

Assuming 8-bit 2's complement numbers.

Sign Extension

- To add two numbers, we must represent them with the **same number** of bits.
- If we just pad with zeros on the left:

<u>4-bit</u>		<u>8-bit</u>	
`0100`	(4)	`00000100`	(still 4)
`1100`	(-4)	`00001100`	(12, not -4)

- Instead, replicate the most significant bit (MSB) -- the sign bit:

<u>4-bit</u>		<u>8-bit</u>	
`0100`	(4)	`00000100`	(still 4)
`1100`	(-4)	`11111100`	(still -4)

What if too big/small?

- Integer and floating-point operations can lead to results too big/small to store within their representations: overflow/underflow

- Binary bit patterns are simply representatives of numbers. Abstraction!
  - Strictly speaking they are called “numerals”.

- Numerals really have an ∞ number of digits
  - with almost all being same (00…0 or 11…1) except for a few of the rightmost digits
  - Just don’t normally show leading digits

- If result of add (or -, *, /) cannot be represented by these rightmost HW bits, we say overflow occurred

Overflow

- Recall the represent range of n-bit 2' complement Signed Integers
- For an n-bit number:

$$-2^{n-1} \sim 2^{n-1} - 1$$

- Can we use n-bit 2' complement to represent a value larger than $2^{n-1}-1$? Or a value smaller than -2^{n-1} ?

Overflow

- If operands are too big, then sum cannot be represented as an *n*-bit 2’ s comp number.

```
  01000  (8)        11000  (-8)
+ 01001  (9)      + 10111  (-9)
  10001  (-15)      01111  (+15)
```

- We have overflow if:
  - signs of both operands are the same, and
  - sign of sum is different.

- Another test -- easy for hardware:
  - carry into MS bit does not equal carry out

```
  01000 (8)         11000 (-8)
+ 01001 (9)       + 10111 (-9)
  10001 (-15)       01111 (+15)
 01                10
```

Logical Operations

- **Operations on logical TRUE or FALSE**
  - two states -- takes one bit to represent:
  - TRUE=1, FALSE=0
- **View *n*-bit number as a collection of *n* logical values**
  - operation applied to each bit independently

A	B	A **AND** B
0	0	0
0	1	0
1	0	0
1	1	1

A	B	A **OR** B
0	0	0
0	1	1
1	0	1
1	1	1

A	**NOT** A
0	1
1	0

Examples of Logical Operations

AND

- useful for clearing bits
  - —AND with zero = 0
  - —AND with one = no change

```
      11000101
AND   00001111
      00000101
```

OR

- useful for setting bits
  - —OR with zero = no change
  - —OR with one = 1

```
      11000101
OR    00001111
      11001111
```

NOT

- unary operation -- one argument
- flips every bit

```
NOT   11000101
      00111010
```

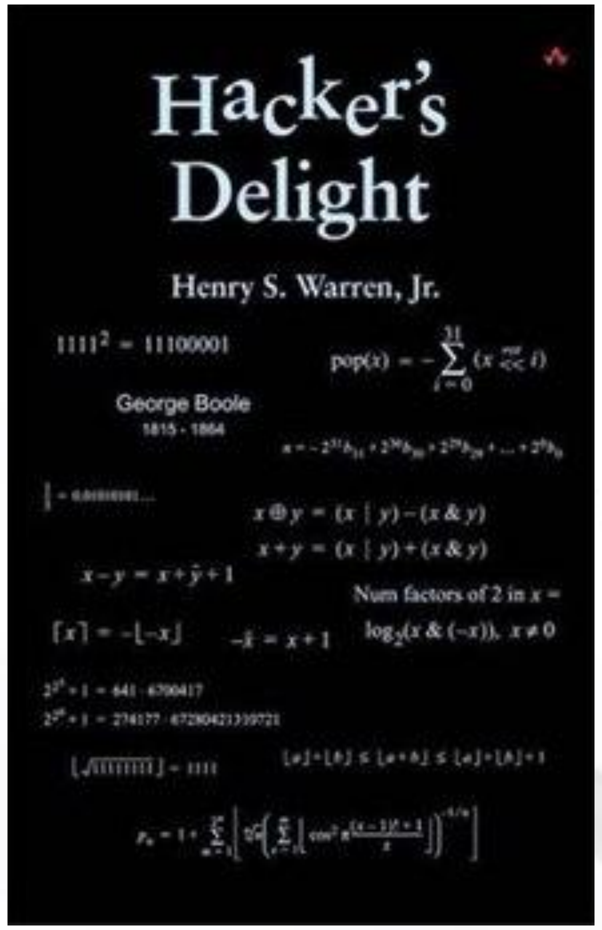
Hacker’s Delight

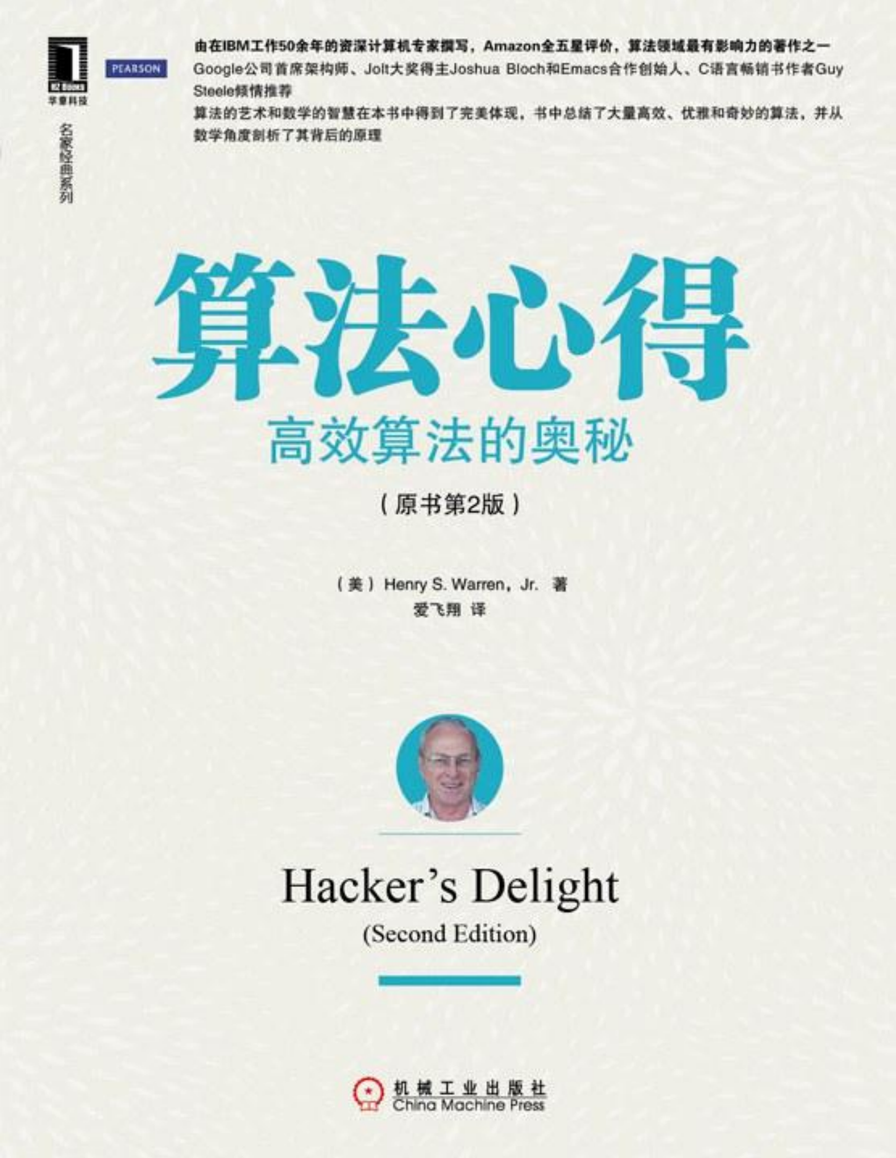
Hacker's Delight

- by Henry S. Warren, Jr.
- first published in 2002
- fast bit-level and low-level arithmetic algorithms

Bit Twiddling Hacks

- By Sean Eron Anderson
- https://graphics.stanford.edu/~seander/bithacks.html





Outline

Fractions: Fixed-Point

- **How can we represent fractions?**
  - Use a "binary point" to separate positive from negative powers of two -- just like "decimal point."

`00101000.101` (40.625)

n	2^n
0	1
1	2
2	4
3	8
4	16
5	32
6	64
7	128
8	256
9	512
10	1024

Fractions: Fixed-Point

■How can we represent fractions?

- Use a "binary point" to separate positive from negative powers of two -- just like "decimal point."
- 2's comp addition and subtraction still work.
  - —if binary points are aligned

$2^{-1} = 0.5$

$2^{-2} = 0.25$

$2^{-3} = 0.125$

```
  00101000.101  (40.625)
+ 11111110.110  (-1.25)
  00100111.011  (39.375)
```

No new operations -- same as integer arithmetic.

n	2^n
0	1
1	2
2	4
3	8
4	16
5	32
6	64
7	128
8	256
9	512
10	1024

Very Large and Very Small Data

- The LC-3 use the 16bit 2’ s complement data type,
- One bit to identify positive or negative, 15bits to represent the magnitude of the value. We can express values:

$$-2^{15} \text{ through } 2^{15}-1$$

(– 32768 through 32767)

How can we represent very large and very small data?

Very Large and Very Small Data

Large values: 6.023×10^{23} — requires **79 bits**

Small values: 6.626×10^{-34} — requires **>110 bits**

How can we represent very large and very small data?

Very Large and Very Small: Floating-Point

Large values: 6.023 x 10^{23} — requires **79 bits**

Small values: 6.626 x 10^{-34} — requires **>110 bits**

Use equivalent of "**scientific notation**": F x 2^E

Need to represent **F** (*fraction/mantissa*), **E** (*exponent*), and **S**(sign).

IEEE 754 Floating-Point Standard (32-bits):

1b	*8b*	*23b*
S	Exponent	Fraction

$$N = (-1)^S \times 1.\text{fraction} \times 2^{\text{exponent}-127}, \quad 1 \le \text{exponent} \le 254$$

$$N = (-1)^S \times 0.\text{fraction} \times 2^{-126}, \quad \text{exponent} = 0$$

Normalized Form

$$N = (-1)^s \times 1.fraction \times 2^{exponent-127},\quad 1 \le exponent \le 254$$

$$exponent: 0b0000\_0000 < exponent < 0b1111\_1111$$

■Single-precision IEEE floating point number:

$$\underbrace{1}_{sign}\ \underbrace{01111110}_{exponent}\ \ \underbrace{10000000000000000000000}_{fraction}$$

- Sign is 1 – number is negative.
- Exponent field, unsigned integer, excess code, biased representations: 01111110 = 126 (decimal).
- Fraction is .100000000000… = .5 (decimal).

Value = -1.5 x $2^{(126-127)}$ = -1.5 x 2^{-1} = -0.75.

Floating Point Example

■Example 2.12

0 01111011 000 0000 0000 0000 0000 0000

\+ 123 0

$$1.0 \times 2^{123-127} = 2^{-4} = \frac{1}{16}$$

■Example 2.13

- $6\frac{5}{8}$
- $(6+\frac{4}{8}+\frac{1}{8})$=-110.101
- 1.10101×2^{2}
- $1.10101\times 2^{129-127}$

1 10000001 101 0100 0000 0000 0000 0000

Floating Point Example

- **Example 2.14**

0 10000011 001 0100 0000 0000 0000 0000

$+\ 1.00101 \times 2^{131-127} = 10010.1 = 18.5$

1 10000010 001 0100 0000 0000 0000 0000

$-\ 1.00101 \times 2^{130-127} = -1001.01 = -9.25$

0 11111110 111 1111 1111 1111 1111 1111

$+\ 1.1111\ldots \times 2^{254-127} = 1.1111\ldots \times 2^{127} \approx 2^{128}$

Very Small: Floating-Point

- **Normalized Form**

$$N = (-1)^s \times 1.fraction \times 2^{exponent-127}, \quad 1 \leq exponent \leq 254$$

$$exponent: 0b0000\_0000 < exponent < 0b1111\_1111$$

- **The smallest number that can be represented in normalized form is**

$$\mathbf{N = 1.00000000000000000000000 \times 2^{-126}}$$

Very Small: subnormal numbers

$$N = (-1)^s \times 0.fraction \times 2^{-126}, \quad exponent = 0$$

- The **largest** subnormal number is

$$\mathbf{N = 0.11111111111111111111111 \times 2^{-126}}$$

- The **smallest** subnormal number is

$$\mathbf{N = 0.00000000000000000000001 \times 2^{-126}}$$
$$\mathbf{= 2^{-23} x 2^{-126} = 2^{-149}}$$

- Example

0 <u>00000000</u> 00001000000000000000000

2^{-5} x 2^{-126} = 2^{-131}

Infinities

■Normalized Form

$$N = (-1)^s \times 1.fraction \times 2^{exponent-127}, \;\; 1 \le exponent \le 254$$

$$exponent: 0b0000\_0000 < exponent < 0b1111\_1111$$

■Subnormal numbers:

$$N = (-1)^s \times 0.fraction \times 2^{-126}, \;\; exponent = 0$$

■So, what if the exponent is equal to 1111_1111?

- If the exponent field contains 1111_1111, we use the floating point data type to represent various things, among them the notion of infinity.
- Infinity is represented by the exponent field containing all 1s and the fraction field containing all 0s.
- We represent positive infinity if the sign bit is 0 and negative infinity if the sign bit is 1

Floating-Point Operations

- **Will regular 2’ s complement arithmetic work for Floating Point numbers?**

(*Hint*: In decimal, how do we compute $3.07 \times 10^{12} + 9.11 \times 10^{8}$?)

Other Data Types

Text strings

- sequence of characters, terminated with NULL (0)
- typically, no hardware support

Image

- array of pixels
  - monochrome: one bit (1/0 = black/white)
  - color: red, green, blue (RGB) components (e.g., 8 bits each)
  - other properties: transparency
- hardware support:
  - typically none, in general-purpose processors
  - MMX -- multiple 8-bit operations on 32-bit word

Sound

- sequence of fixed-point numbers

Within the Computer: Everything is a Number.

How do computers represent text ? -- ASCII Characters

ASCII: Maps 128 characters to 7-bit code.

- both printable and non-printable (ESC, DEL, …) characters

00	nul	10	dle	20	sp	30	0	40	@	50	P	60	`	70	p
01	soh	11	dc1	21	!	31	1	41	A	51	Q	61	a	71	q
02	stx	12	dc2	22	"	32	2	42	B	52	R	62	b	72	r
03	etx	13	dc3	23	#	33	3	43	C	53	S	63	c	73	s
04	eot	14	dc4	24	$	34	4	44	D	54	T	64	d	74	t
05	enq	15	nak	25	%	35	5	45	E	55	U	65	e	75	u
06	ack	16	syn	26	&	36	6	46	F	56	V	66	f	76	v
07	bel	17	etb	27	'	37	7	47	G	57	W	67	g	77	w
08	bs	18	can	28	(	38	8	48	H	58	X	68	h	78	x
09	ht	19	em	29	)	39	9	49	I	59	Y	69	i	79	y
0a	nl	1a	sub	2a	*	3a	:	4a	J	5a	Z	6a	j	7a	z
0b	vt	1b	esc	2b	+	3b	;	4b	K	5b	[	6b	k	7b	{
0c	np	1c	fs	2c	,	3c	<	4c	L	5c	\	6c	l	7c	|
0d	cr	1d	gs	2d	-	3d	=	4d	M	5d	]	6d	m	7d	}
0e	so	1e	rs	2e	.	3e	>	4e	N	5e	^	6e	n	7e	~
0f	si	1f	us	2f	/	3f	?	4f	O	5f	_	6f	o	7f	del

ASCII (American Standard Code for Information Interchange)

ASCII表

(American Standard Code for Information Interchange 美国标准信息交换代码)

高四位		ASCII控制字符												ASCII打印字符												
		0000						0001						0010		0011		0100		0101		0110		0111		
		0						1						2		3		4		5		6		7		
低四位		十进制	字符	Ctrl	代码	转义字符	字符解释	十进制	字符	Ctrl	代码	转义字符	字符解释	十进制	字符	十进制	字符	十进制	字符	十进制	字符	十进制	字符	十进制	字符	Ctrl
0000	0	0		^@	NUL	\0	空字符	16	►	^P	DLE		数据链路转义	32		48	0	64	@	80	P	96	`	112	p	
0001	1	1	☺	^A	SOH		标题开始	17	◄	^Q	DC1		设备控制 1	33	!	49	1	65	A	81	Q	97	a	113	q	
0010	2	2	☻	^B	STX		正文开始	18	↕	^R	DC2		设备控制 2	34	"	50	2	66	B	82	R	98	b	114	r	
0011	3	3	♥	^C	ETX		正文结束	19	‼	^S	DC3		设备控制 3	35	#	51	3	67	C	83	S	99	c	115	s	
0100	4	4	♦	^D	EOT		传输结束	20	¶	^T	DC4		设备控制 4	36	$	52	4	68	D	84	T	100	d	116	t	
0101	5	5	♣	^E	ENQ		查询	21	§	^U	NAK		否定应答	37	%	53	5	69	E	85	U	101	e	117	u	
0110	6	6	♠	^F	ACK		肯定应答	22	▬	^V	SYN		同步空闲	38	&	54	6	70	F	86	V	102	f	118	v	
0111	7	7	•	^G	BEL	\a	响铃	23	↨	^W	ETB		传输块结束	39	'	55	7	71	G	87	W	103	g	119	w	
1000	8	8	◘	^H	BS	\b	退格	24	↑	^X	CAN		取消	40	(	56	8	72	H	88	X	104	h	120	x	
1001	9	9	○	^I	HT	\t	横向制表	25	↓	^Y	EM		介质结束	41	)	57	9	73	I	89	Y	105	i	121	y	
1010	A	10	◙	^J	LF	\n	换行	26	→	^Z	SUB		替代	42	*	58	:	74	J	90	Z	106	j	122	z	
1011	B	11	♂	^K	VT	\v	纵向制表	27	←	^[	ESC	\e	溢出	43	+	59	;	75	K	91	[	107	k	123	{	
1100	C	12	♀	^L	FF	\f	换页	28	∟	^\	FS		文件分隔符	44	,	60	<	76	L	92	\	108	l	124	|	
1101	D	13	♪	^M	CR	\r	回车	29	↔	^]	GS		组分隔符	45	-	61	=	77	M	93	]	109	m	125	}	
1110	E	14	♫	^N	SO		移出	30	▲	^^	RS		记录分隔符	46	.	62	>	78	N	94	^	110	n	126	~	
1111	F	15	☼	^O	SI		移入	31	▼	^-	US		单元分隔符	47	/	63	?	79	O	95	_	111	o	127	⌂	^Backspace 代码：DEL

注：表中的ASCII字符可以用“Alt + 小键盘上的数字键”方法输入。

Interesting Properties of ASCII Code

- What is relationship between a decimal digit ('0', '1', …)and its ASCII code?
- What is the difference between an upper-case letter ('A', 'B', …) and its lower-case equivalent ('a', 'b', …)?
- Given two ASCII characters, how do we tell which comes first in alphabetical order?
- Are 128 characters enough? (http://www.unicode.org/)

No new operations -- integer arithmetic and logic.

How do computers represent image ?

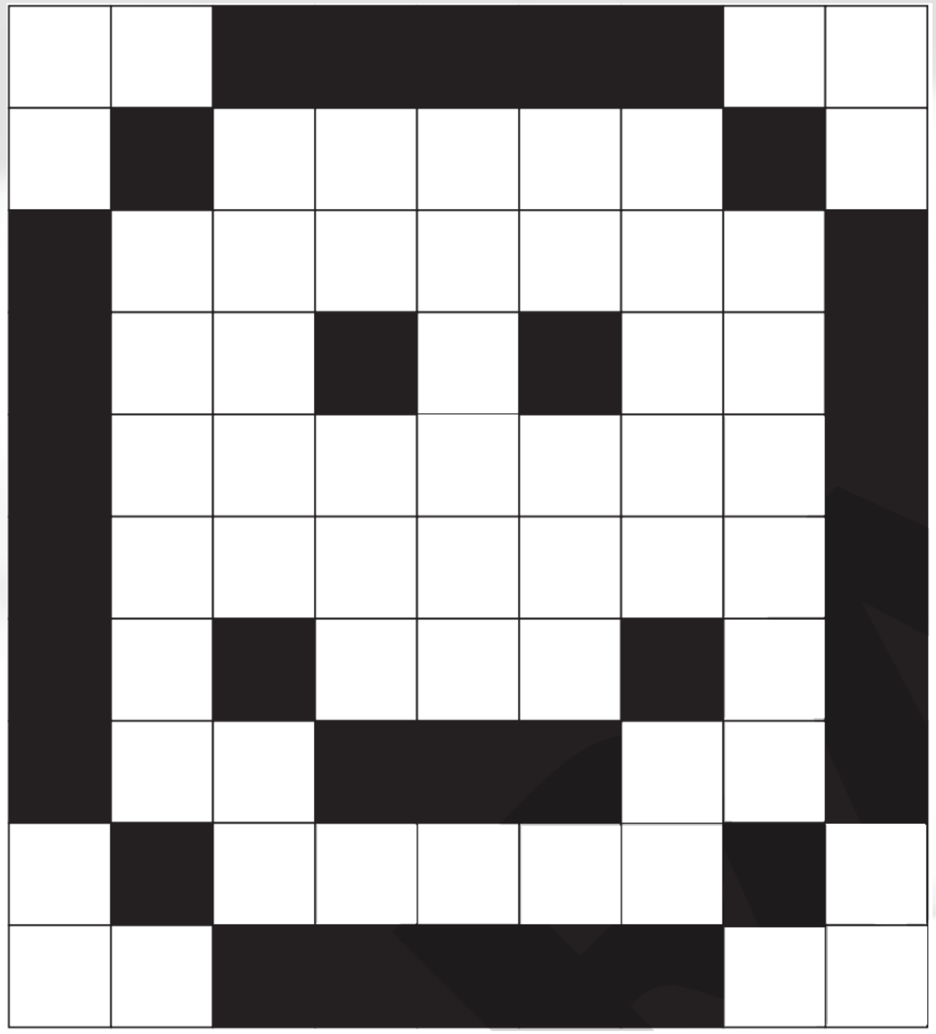
- Each image has a **resolution** and a **color depth.**
  - The **resolution** is the number of pixels wide and the number of pixels high that are used to create the image.
  - The **color depth** is the number of bits that are used to represent each color.

Figure 1.3: 8-bit colour 16-bit colour 32-bit colour

- For example, each color could be represented using 8-bit, 16-bit or 32-bit binary numbers.
- The greater the number of bits, the greater the range of colors that can be represented.

Converting images to binary

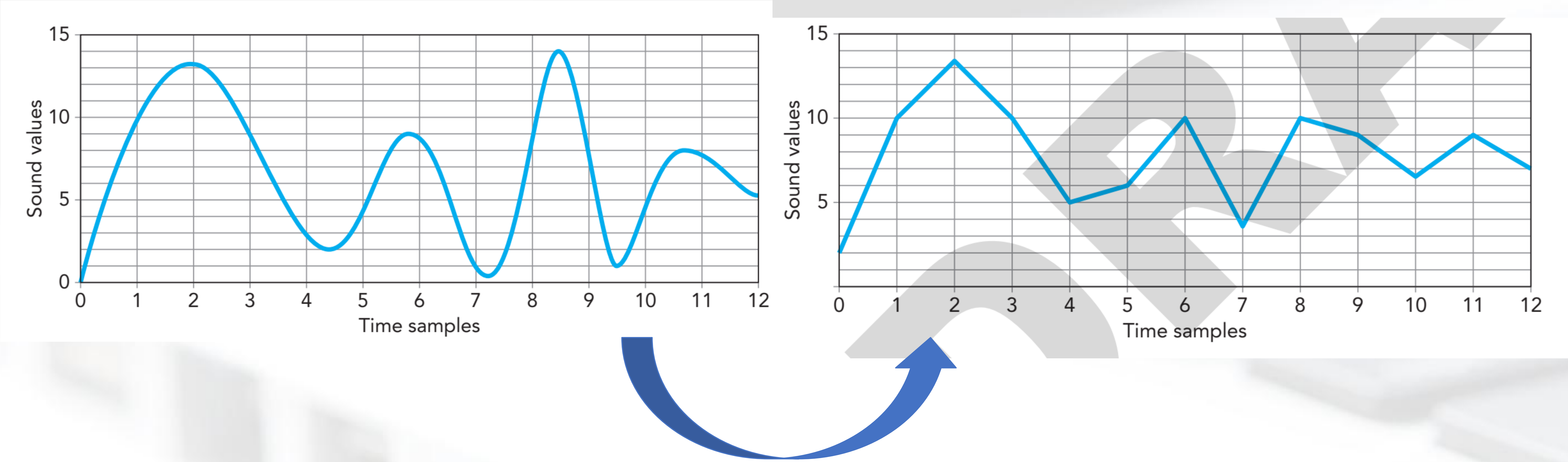
- If each pixel is converted to its binary value, a data set such as the following could be created:



0	0	1	1	1	1	1	0	0
0	1	0	0	0	0	0	1	0
1	0	0	0	0	0	0	0	1
1	0	0	1	0	1	0	0	1
1	0	0	0	0	0	0	0	1
1	0	1	0	0	0	1	0	1
1	0	0	1	1	1	0	0	1
0	1	0	0	0	0	0	1	0
0	0	1	1	1	1	1	0	0

How do computers represent sound ?

- Sound is made up of sound waves. When sound is recorded, this is done at set time intervals. This process is known as **sound sampling** :



Time sample	1	2	3	4	5	6	7	8	9	10	11	12
Sound value	9	13	9	3.5	4	9	1.5	9	8	5	8	5.5

LC-3 Data Types

- Some data types are supported directly by the instruction set architecture.
- For LC-3, there is only one supported data type:
  - 16-bit 2's complement signed integer
  - Operations: ADD, AND, NOT
- Other data types are supported by interpreting 16-bit values as logical, text, fixed-point, etc., in the software that we write.